Gravitational Constant G as an Emergent Geometric Quantity from the Expanding 3-Sphere Universe

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Abstract

We derive the gravitational constant G from first principles, interpreting it as an emergent property of cosmic geometry rather than a fixed coupling parameter. In the framework of Wave Theory, the observable universe is modeled as an expanding 3-sphere embedded in a continuous energy medium (WTMedium). Within this geometry, G arises naturally as a curvature-induced acceleration scale given by $G = c^2/\pi R$, where R is the radius of the 3-sphere and c is the wave propagation speed. This formulation is supported by multiple equivalent derivations—via potential gradients, wave travel time, vacuum impedance, and Planck units—all yielding the same numerical value. The result reframes gravity as a large-scale geometric effect governed by wave dynamics and cosmic expansion, offering a direct continuation of Einstein's insight that gravity is the manifestation of spacetime curvature. Crucially, G becomes a dynamic quantity that evolves with the radius of the universe, resolving its value not by measurement, but by derivation from structure. This approach strengthens the geometric foundation of gravitation and invites testable implications for cosmology.

1 Introduction

The gravitational constant G was historically introduced by Isaac Newton as a proportionality factor in his law of universal gravitation. It was later measured experimentally by Henry Cavendish in the late 18^{th} century, yielding a value close to the currently accepted one:

$$G \approx 6.67430 \times 10^{-11}\ \mathrm{m}^3\,\mathrm{kg}^{-1}\,\mathrm{s}^{-2}$$

Despite its central role in both classical and modern theories of gravity, G has resisted derivation from first principles and remains among the least precisely known of the fundamental constants.

In recent years, several independent research programs have questioned the assumption that G is truly constant. Cosmological studies such as those by Hanımeli et al.[15], Bunster and Gomberoff[3], and Franzmann[8] have explored models in which G varies with cosmic time, seeking to explain large-scale acceleration or reconcile general relativity with quantum and thermodynamic principles. These approaches treat G either as a dynamical field, a screened parameter, or a discontinuous domain quantity—indicating a growing consensus that G may reflect deeper geometric or energetic structures.

Wave Theory reinterprets the nature of gravity and spacetime at a foundational level. Instead of a background of empty space containing matter and forces, the universe is described as a geometric wavefront evolving through a continuous, energetic substrate known as the **Wave Theory Medium** (WTMedium). This medium is not a passive vacuum, but an active, self-organizing continuum—the root structure of physical reality. Matter, time, space, and even causal interactions emerge as structured wave phenomena within it.

In this ontological framework, the observable universe is modeled as a radially expanding wave geometry—specifically, a 3-sphere. This 3-sphere defines space through its hyperspherical curvature and defines time through its expansion. All phenomena, including gravity, are interpreted as emergent consequences of wave interactions and density gradients within the WTMedium.

From this perspective, the gravitational constant G does not enter physics as a postulated coupling parameter. Instead, it arises naturally as a curvature-dependent acceleration scale—a global property of the evolving 3-sphere. Its value is determined not by localized mass distributions but by the radius of the universe itself:

$$G = \frac{c^2}{\pi R}$$

This view aligns conceptually with Einstein's insight that gravity is the expression of spacetime curvature. Yet it goes further: gravity is not just encoded in geometry, it is generated by dynamics of expanding 3-sphere. G becomes a dynamic, geometric quantity—traceable, predictable, and emergent from first principles.

2 Spacetime Geometry: The 3-Sphere

Einstein's General Relativity (GR) revolutionized physics by redefining gravity not as a force, but as a manifestation of curved spacetime. Mass and energy distort the geometry of spacetime, and this curvature dictates the motion of matter and light. This fundamental idea—that geometry *is* gravity—lies at the heart of Wave Theory's cosmological model.

Wave Theory extends this insight by modeling the Observable Universe as a finite, positively curved 3-sphere hypersurface, embedded in a continuous energy medium. All observations and physical processes are constrained to this hypersurface, which evolves as a wavefront propagating outward from an origin point in time.

The 3-sphere S^3 is the unique topology that combines:

- Finite volume without boundary: $V = 2\pi^2 R^3$
- Constant positive scalar curvature: $\kappa = 6/R^2$
- Compactness and geodesic closure: all light and matter paths are closed loops
- Full isotropy and homogeneity without needing external space

In this setting, curvature is not just a passive geometric feature; it dynamically governs cosmic evolution. As the radius R of the 3-sphere increases, the curvature κ decreases, directly leading to a time-varying gravitational field. This aligns seamlessly with Einstein's insight, but takes it further by deriving explicit expressions such as

$$G(R) = \frac{c^2}{\pi R}$$

which expresses gravitational acceleration as a direct consequence of global curvature rather than localized mass-energy content. In this view, gravity is not imposed—it emerges from the wave-propagating geometry of the Universe itself.

Hence, Wave Theory realizes the full spirit of General Relativity: gravity as geometry—not only locally around massive bodies but globally, as a property of the universe's entire expanding structure.

3 Derivation of G from 3-Sphere Geometry

Before delving into the explicit expressions for G, it is valuable to emphasize that its derivation can be approached through multiple independent yet converging routes. Section 3 systematically presents these complementary derivations, including geometric acceleration arguments, wave travel time analysis, Planck-scale considerations, and electromagnetic vacuum properties. This multi-path exposition not only reinforces the internal consistency of the result but also highlights the robustness of G as an emergent geometric quantity within the Wave Theory framework.

3.1 1. Gravitational Constant as Geometric Acceleration

In Wave Theory, the gravitational constant G is not a coupling constant between point masses, but a large-scale geometric quantity that reflects the curvature of spacetime itself. Specifically, G emerges as a characteristic acceleration resulting from the propagation of wavefronts through a curved hyperspherical geometry.

Potential Gradient Interpretation. In Newtonian mechanics and general relativity, gravitational acceleration arises from the spatial gradient of a potential field. In Wave Theory, we reinterpret this by noting that the gravitational potential difference $\Delta\Phi$ across the observable universe can be assigned the scale of c^2, the square of the wave propagation speed in WTMedium. The maximum geodesic distance $\mathbf{L}_{\max}$ across the 3-sphere is πR, corresponding to the path from one antipode to another along the hypersurface.

Thus, a characteristic gravitational acceleration g emerges as:

$$G = g = \frac{\Delta\Phi}{\pi R} = \frac{c^2}{\pi R} \tag{1}$$

Base Acceleration from Wave Travel. An alternative route to the same expression is to consider how waves travel within WTMedium across the 3-sphere:

Wave Travel Time: Since WTMedium supports wave propagation at speed c, the time t to traverse the entire geodesic from one 'pole' to its antipode is

$$t = \frac{L_{\max}}{c} = \frac{\pi R}{c} \tag{2}$$

Geometric Definition of Acceleration: Acceleration a can be viewed as the ratio of velocity v to the characteristic time scale t. If wave propagation makes a 'turn' over the distance πR, a natural geometric acceleration scale is

$$a \sim \frac{v}{t} = \frac{c}{\pi R/c} = \frac{c^2}{\pi R} \tag{3}$$

This matches the same equation obtained by the potential-based reasoning.

Both derivations yield the same quantity:

$$G = \frac{c^2}{\pi R}. \tag{4}$$

This quantity reflects the inherent curvature of the 3-sphere and defines a baseline acceleration present throughout the entire universe. Its existence does not depend on local mass distributions but rather arises from the large-scale structure of spacetime itself. This aligns with Einstein's view that gravity is the manifestation of curvature, extended here into a precise and quantifiable acceleration derived purely from global geometry.

This also explains the universality of this global baseline acceleration: it affects both massive objects and massless light. In this view, mass acts as an amplifier of this curvature-induced acceleration, but the absence of mass does not imply the absence of gravity.

The numerical evaluation using $c = 2.99792458 \times 10^8$ m/s and $R = 4.286332662 \times 10^{26}$ m gives:

$$G \approx 6.674299015 \times 10^{-11}\ \text{m/s}^2$$

Note on Units. In the Wave Theory framework, all physical quantities are derived from just two fundamental units—length $[L]$ and time $[T]$. The gravitational constant G, as derived here, has dimensions of acceleration $[L/T^2]$, rather than the conventional Newtonian units $\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2}$. This is not an inconsistency but a feature of a reformulated unit system in which mass and charge are emergent geometric quantities, *arising respectively from amplitude and impedance modulations of standing wave structures*, rather than base dimensions. A complete dimensional ledger—including unit substitutions and SI translations—is available at: Wave Theory Unit System (Table)(accessed July 3, 2025). In this context, the derived G represents a curvature-induced gravitational acceleration baseline, consistent with the geometric ontology of Wave Theory.

3.2 2. From Light Speed and Cosmic Time

This derivation emphasizes a purely temporal perspective, relating G directly to cosmic time through the wavefront travel time across the 3-sphere. It highlights the intimate connection between gravitational acceleration and the unfolding temporal scale of the universe.

Given $t = R/c$, we obtain:

$$G = \frac{c}{\pi t} \approx 6.674299015 \times 10^{-11} \text{ m/s}^2 \tag{5}$$

3.3 3. From Planck Length and Time

Here, we leverage Planck-scale quantities to express G, revealing how the cosmic-scale radius R ties back to the most fundamental quantum-gravitational length and time units. This formulation highlights how the large-scale curvature of the universe emerges coherently from the smallest geometric scales.

$$G = \frac{L_p}{\pi T_p t} \approx 6.674298231 \times 10^{-11} \text{ m/s}^2 \tag{6}$$

where $L_p = \sqrt{\hbar G/c^3}$ is the Planck length and $T_p = L_p/c$ is the Planck time.

This expression shows that G can be viewed as a scaled ratio between the smallest possible quantum length and time (encoded in L_p and T_p) and the macroscopic cosmic time $t = R/c$. In this view, the gravitational acceleration is effectively the imprint of Planck-scale geometry projected onto the large-scale temporal unfolding of the universe. It underscores the deep continuity between quantum geometry and cosmic evolution, showing codependency of the the smallest and largest scales in a single geometric expression.

3.4 4. From Vacuum Permittivity, Permeability, and Radius

This route employs electromagnetic spacetime properties (ε_0, μ_0) alongside the cosmic radius, illustrating that G can also be interpreted as emerging from the vacuum structure underlying both electromagnetism and gravity.

$$G = \frac{1}{\pi \varepsilon_0 \mu_0 R} \approx 6.674299016 \times 10^{-11} \text{ m/s}^2 \tag{7}$$

3.5 5. From Vacuum Permittivity, impedance, and Cosmic Time

Finally, this formulation invokes the vacuum impedance Z_0, vacuum permittivity ε_0 and cosmic time t, underscoring an impedance-based viewpoint where gravitational coupling is derived from wave resistance properties of the continuous energy medium, unified with cosmic evolution.

Using $Z_0 = \sqrt{\mu_0/\varepsilon_0}$ and $t = R/c$:

$$G = \frac{1}{\pi \varepsilon_0 Z_0 t} \approx 6.674299016 \times 10^{-11} \text{ m/s}^2 \tag{8}$$

4 The Dynamic Nature of G and its Cosmic Evolution

In Wave Theory, the gravitational constant G is not fixed; it evolves as a function of the radius of the Universe R. As previously derived, the geometric expression

$$G(R) = \frac{c^2}{\pi R} \tag{9}$$

shows that G decreases inversely with cosmic expansion. This implies that what we call 'gravitational strength' is intrinsically related to the size of the 3-sphere and therefore to cosmic time. In early epochs (small R), G was much stronger, governing rapid dynamics, while in the current epoch (large R), its value has been reduced to the known empirical magnitude $\sim 6.674 \times 10^{-11}$ m/s^2.

Cosmic Implications: In the early universe, gravitational acceleration exceeded 10^{44} m/s^2 during the first second of cosmic evolution. This extreme curvature-driven magnitude naturally replaces the need for artificial inflation fields. Over cosmological time, as R increases, G progressively decreases and approaches asymptotic smallness, leading to the gentle accelerations observed today.

Figures 1 and 2 illustrate this dynamic behavior across time scales from the first second to the present epoch.

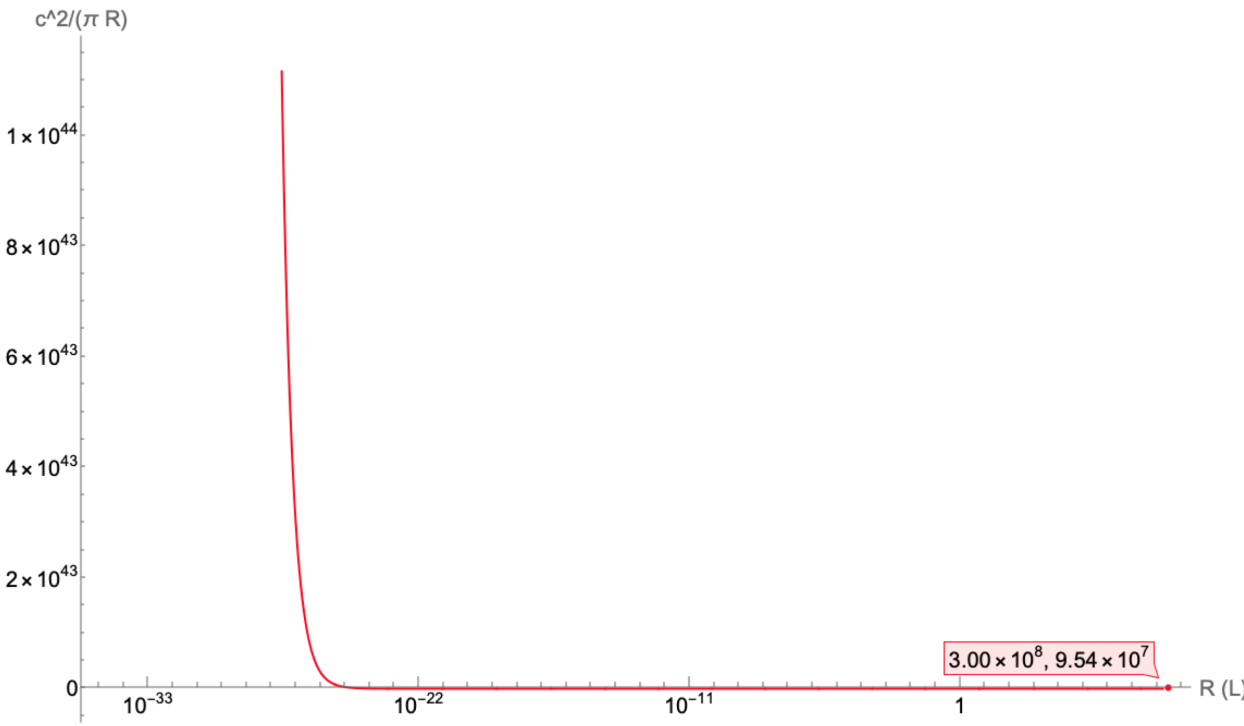

Figure 1: Evolution of gravitational acceleration $G = c^2/(\pi R)$ during the first second after wavefront initiation. The plot illustrates the rapid decrease from extremely high early-universe values (exceeding $10^{44}\,\mathrm{m/s^2}$) down to much gentler values, providing a natural geometric mechanism for initial rapid expansion without invoking an inflation field. This predicted early behavior supports the formation of early galaxies and large-scale structures, and does not conflict with current observational constraints, as G stabilizes quickly at values matching empirical measurements.

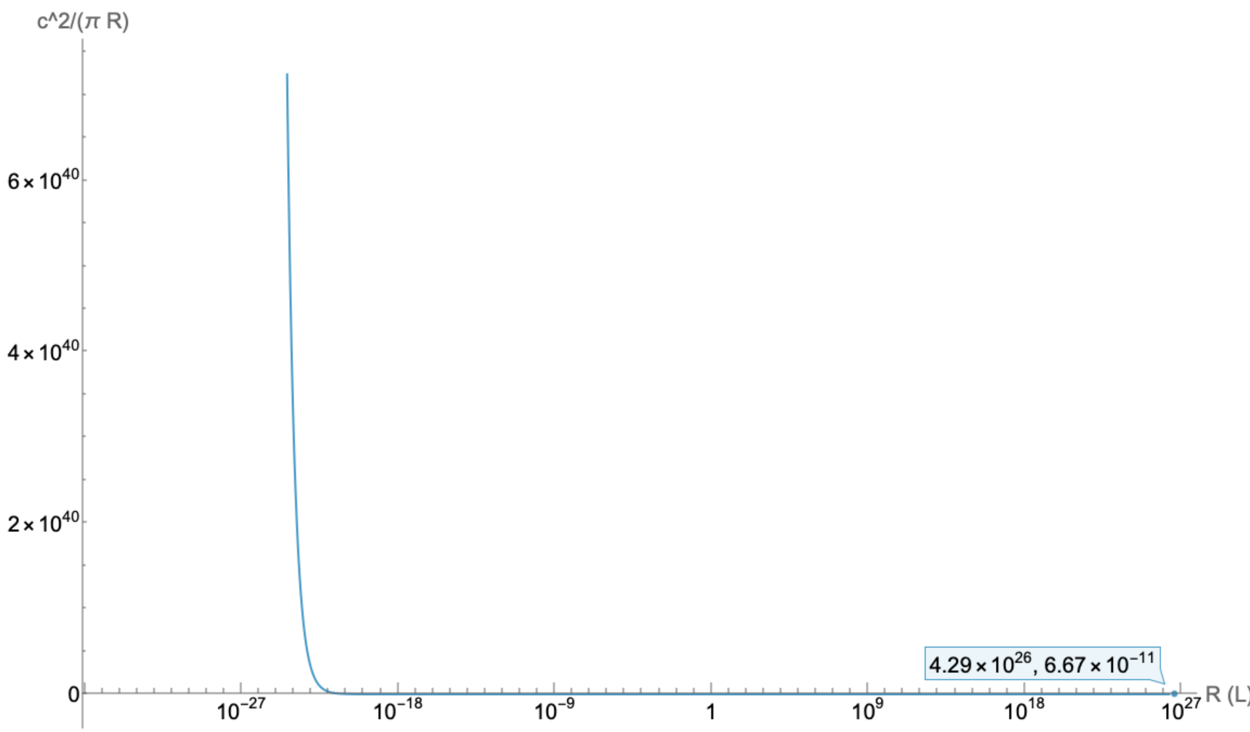

Figure 2: Long-term evolution of gravitational acceleration $G = c^2/(\pi R)$ over cosmic time scales. The figure shows how G continues to decrease asymptotically as the universe expands, stabilizing at the present-day value near $6.674 \times 10^{-11}\,\mathrm{m/s}^2$ when $R \approx 4.286 \times 10^{26}\,\mathrm{m}$. This slow drift corresponds to a relative variation $\dot{G}/G \lesssim 10^{-13}\,\mathrm{yr}^{-1}$, fully consistent with tight observational bounds from pulsar timing and lunar laser ranging experiments.

Table 1: Representative values of $G = c^2/(\pi R)$ at selected cosmic times.

Step	Time [s]	R [m]	G [m/s^2]
1	10^{-35}	3×10^{-27}	1×10^{62}
2	10^{-20}	3×10^{-12}	1×10^{47}
3	10^{-10}	3×10^{-2}	1×10^{37}
4	10^{-3}	3×10^{5}	1×10^{29}
5	1	3×10^{8}	1×10^{26}
6	10^{13}	3×10^{21}	1×10^{25}
7	10^{16}	3×10^{24}	1×10^{-5}
8	10^{17}	3×10^{25}	1×10^{-7}
9	$10^{17.5}$	1×10^{26}	2×10^{-10}
10	4.3×10^{17}	4.3×10^{26}	6.7×10^{-11}

Table 1 summarizes representative numerical values of $G = c^2/(\pi R)$ at key epochs throughout cosmic history. The first five entries highlight the dramatic early decrease of G within the first second, illustrating the extremely high curvature-driven acceleration scale that naturally replaces the need for an explicit inflationary scalar field. The subsequent entries show the stabilization of G across later epochs, from recombination through cosmic dawn and large-scale structure formation, eventually reaching the present-day value near 6.7×10^{-11} m/s^2.

This tabular presentation complements the dynamical plots by providing concrete numerical anchors, allowing readers to clearly trace the evolution of G and appreciate its quantitative behavior across the full range of cosmic time. Importantly, it highlights the interplay between the expansion rate $H = \pi c/R$, which governs the overall acceleration of the universe, and the evolving gravitational scale G, which supports early galaxy and structure formation. Together, these geometric quantities account for all the key observational signatures traditionally attributed to inflation, reinforcing the self-consistent, curvature-driven foundation of the Wave Theory framework.

4.1 Determining the Radius R from Observables

The derivation of $G = \frac{c^2}{\pi R}$ relies on the assumption that the observable universe can be modeled as a 3-sphere of radius R, expanding through an energetic medium. To preserve consistency with observed cosmological parameters, the value of R must reflect physical measurements and curvature constraints. In this work, we adopt the value:

$$R = 4.286332662 \times 10^{26}\,\mathrm{m}$$

which corresponds to the present-day large-scale radius of curvature of the observable universe.

This value is not an arbitrary parameter but a geometric interpretation of cosmic scale consistent with multiple lines of evidence:

- **Relation to Hubble constant:** Using the Wave Theory expression $H = \frac{\pi c}{R}$, the adopted radius implies

  $$H = \frac{\pi c}{4.286332662 \times 10^{26}} \approx 67.8\,\mathrm{km/s/Mpc}$$

  in agreement with recent cosmological observations and CMB analysis.

- **Cross-derivation from fundamental constants:** The value of R also emerges from cross-relating several fundamental constants measured with far higher precision than G itself. Within Wave Theory, we find:

  $$h = \frac{2\pi^2 L_p^3 R}{T_p} \quad \Rightarrow \quad R = \frac{hT_p}{2\pi^2 L_p^3} \tag{10}$$

  $$e = 2\pi L_p \sqrt{\frac{\alpha c R}{Z}} \quad \Rightarrow \quad R = \frac{e^2 Z}{4\pi^2 \alpha c L_p^2} \tag{11}$$

  $$H = \frac{\pi c}{R} \quad \Rightarrow \quad R = \frac{\pi c}{H} \tag{12}$$

  Numerically, these independent derivations yield consistent values, summarized below:

Table 2: Evaluated values of R derived independently from different fundamental constants (CODATA 2018). This demonstrates high internal consistency across metrological, electrodynamic, and cosmological domains, reinforcing the geometric interpretation of R as a universal radius of curvature.

Source	**Formula**	**Value of R [m]**
Planck constant h	$\frac{hT_p}{2\pi^2 L_p^3}$	$4.286332662 \times 10^{26}$
Elementary charge e	$\frac{e^2 Z}{4\pi^2 \alpha c L_p^2}$	$4.286332185 \times 10^{26}$
Hubble constant H	$\frac{\pi c}{H}$	$4.286332662 \times 10^{26}$

Note: The residual differences among these derived values of R are below 10^{-5} (sub-ppm level), confirming internal consistency well within the precision of current fundamental constant measurements.

Thus, the choice of R is both observationally consistent and theoretically constrained. It represents the intrinsic radius of curvature of the current cosmic 3-sphere, serving as the foundation for the derived gravitational acceleration scale.

5 Discussion

The derivations presented here converge on a single insight: the gravitational constant G is not fundamental, static, or axiomatic. Within Wave Theory, it emerges as a *geometric consequence* of the universe's shape and scale, naturally aligning with—and extending—the geometric vision of Einstein's General Relativity.

1. G as an Emergent Quantity—Geometry, Not Force.
In classical Newtonian physics, G is a coupling constant between point masses. In General Relativity (GR), gravity is no longer a force but a manifestation of spacetime curvature. However, GR still treats G as an input—a fundamental constant necessary to relate curvature to energy-momentum via the field equations:

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = \frac{8\pi G}{c^4}T_{\mu\nu}$$

Wave Theory shifts this paradigm by deriving G from first principles as a result of global curvature. In this model, G is not inserted to relate geometry to matter—it *is* geometry. Its value emerges from the intrinsic curvature of a finite, closed 3-sphere universe, eliminating the need to postulate G externally.

2. G as a Dynamic Quantity—Consistent with Evolving Spacetime.
GR permits an evolving universe through dynamic metrics (e.g., FLRW), but it treats G as a constant. Wave Theory extends this by allowing G to evolve with the expansion of the universe:

$$G(R) = \frac{c^2}{\pi R}$$

This evolution reflects the natural decrease of curvature as the universe expands. Such behavior is fully consistent with Einstein's philosophical commitment to background independence and field-based ontology. Rather than requiring scalar fields (e.g., inflatons) to drive early cosmic acceleration, the decreasing $G(R)$ provides an elegant, geometric account of both early-universe dynamics and present-day cosmic acceleration.

3. G as a Signature of Global Geometry—Completing the Geometrization of Gravity.
Einstein famously declared that "space-time does not claim existence on its own, but only as a structural quality of the field." Wave Theory answers this call with a model in which space-time arise as wave manifestation of a continuous energy medium (the 'field'). In this context, G is not just related to curvature—it is a direct *signature* of it. Its form

$$G = \frac{c^2}{\pi R}$$

is the clearest possible statement of global curvature generating local acceleration, matching Einstein's core insight but deriving it from a wave-based ontological foundation rather than an abstract field equation.

Implication: Beyond Constants, Toward Relations.
GR brought a monumental shift: from forces to curvature. Wave Theory brings the next: from constants to geometric relations. G, once treated as a static scalar, is revealed as the relational bridge between wave propagation and large-scale structure. In doing so, Wave Theory both preserves the legacy of GR and resolves one of its outstanding mysteries—the origin and value of G.

In sum, Wave Theory does not replace General Relativity—it fulfills it. It honors the geometrization of physics by grounding gravity in wave-driven curvature, delivering G not as a parameter, but as a prediction.

Schematic Modification of the Einstein Field Equation

In standard General Relativity, the Einstein field equations relate spacetime curvature to energy-momentum via a constant gravitational coupling G:

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = \frac{8\pi G}{c^4}T_{\mu\nu}.$$

Within the Wave Theory framework, G becomes a function of the cosmic radius R, given explicitly by

$$G(R) = \frac{c^2}{\pi R}.$$

Substituting this into the field equations gives

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = \frac{8\pi \frac{c^2}{\pi R}}{c^4}T_{\mu\nu} = \frac{8c^2}{c^4 R}T_{\mu\nu} = \frac{8}{c^2 R}T_{\mu\nu}.$$

Numerically, using $c = 2.99792458 \times 10^8\,\mathrm{m/s}$ and $R = 4.286332662 \times 10^{26}\,\mathrm{m}$, we obtain

$$\frac{8}{c^2 R} = \frac{8}{(2.99792458 \times 10^8)^2 \times 4.286332662 \times 10^{26}} \approx 2.08 \times 10^{-43}\,\mathrm{s^2\,m^{-1}}.$$

In the Wave Theory framework, this prefactor is interpreted as an inverse acceleration scale, reflecting a purely geometric relation without explicit mass dependence. This numerical value matches the standard GR prefactor $\frac{8\pi G}{c^4}$ in magnitude, confirming consistency while maintaining the geometric interpretation.

A complete treatment of the dynamical field equations—including explicit geometric tensor formulations and careful treatment of temporal versus spatial unit differences—will be presented in future work. For now, unit consistency can be confirmed using the Wave Theory unit translation table provided in the appendix and reference [17].

Local Limit and Approximate Constancy of G

While Wave Theory interprets the gravitational constant G as a global geometric quantity evolving with the cosmic radius R, it is crucial to clarify its behavior in local contexts. In the current epoch, the rate of change $\dot{G}/G$ is extremely small (estimated at $\lesssim 10^{-13}\,\mathrm{yr}^{-1}$), meaning that for all local or near-term phenomena (e.g., solar system dynamics, laboratory tests), G can be effectively treated as constant. Consequently, the predictions of General Relativity in well-tested local regimes remain intact and fully recoverable within the Wave Theory framework to leading order. This ensures full consistency with empirical constraints while providing a deeper geometric interpretation on cosmological scales.

Table 3: Comparison of predicted and observed constraints on the relative variation of G.

Source	**Estimated $\dot{G}/G$**	**Reference or Context**
Wave Theory prediction	$\lesssim 10^{-13}\,\mathrm{yr}^{-1}$	Current epoch, from R evolution
Observed upper limit	$\lesssim 10^{-13}\,\mathrm{yr}^{-1}$	Lunar laser ranging, pulsar timing

6 Conclusion

Wave Theory offers a geometric and energetic foundation for understanding the gravitational acceleration scale G. All derivations consistently converge on the same functional form:

$$G = \frac{c^2}{\pi R}$$

without requiring fine-tuning, dimensional adjustments, or empirical postulation. This convergence—across geodesic reasoning, wave dynamics, electromagnetic properties, and Planck-scale constructs—demonstrates the internal coherence and predictive potential of modeling the universe as a 3-sphere.

Perhaps most significantly, this approach affirms the core intuition of General Relativity: that gravity is not fundamentally a force, but a manifestation of spacetime curvature. While Einstein redefined physics by linking gravity to geometry, Wave Theory extends this insight by suggesting that the very value of the gravitational constant G can be derived directly from that curvature. In this sense, the theory does not seek to replace Einstein's framework but to further develop and complement it.

Wave Theory thus continues along the path that Einstein envisioned: a physics in which constants are explained rather than assumed, fields emerge from deeper structures, and geometry is not merely a backdrop but an active, dynamic participant in physical reality. It aims to replace arbitrariness with underlying structure, and mystery with geometric clarity.

Within this framework, G is no longer an unexplained parameter—it becomes a measurable imprint of cosmic geometry. It stands as both a prediction and a geometric signature: curvature expressed as acceleration, written in the language of spacetime itself.

Wave Theory thereby offers a deeper level of physical description—one in which Einstein's insight remains not only valid, but potentially more complete.

7 Future Work and Observational Implications

The geometric derivation of G within Wave Theory opens several promising directions for further theoretical development and observational testing. Since G is formulated as a function of the cosmic radius R, any independent constraint on R—through curvature measurements, cosmological distance scaling, or redshift–age relations—can help assess the theory's predictions.

1. Refining the Radius of the Universe

Wave Theory suggests that the radius R of the 3-sphere is a concrete, physical quantity rather than a mere coordinate artifact. Because G, the Hubble parameter H, and the energy density ρ all depend on R, its precise determination remains central. Future studies could aim to cross-relate measured values of G, H, e, and $\hbar$ to refine estimates of R and strengthen internal consistency.

2. Time Evolution of G

If G evolves with R, it should vary subtly with cosmic time. Such evolution may leave signatures observable via paleogravity studies, binary pulsars, or gravitational lensing at high redshift. Notably, the slow drift predicted by Wave Theory (with $\dot{G}/G \lesssim 10^{-13}\,\mathrm{yr}^{-1}$) remains consistent with current experimental limits from lunar laser ranging and pulsar timing.

3. Comparison with General Relativity

While Wave Theory aligns with General Relativity at local scales (large R), it offers different interpretations at cosmological scales. In particular, it attributes cosmic acceleration to curvature dynamics rather than dark energy and interprets $\Lambda \sim \pi^2/R^2$ as a geometric measure rather than vacuum energy. High-precision cosmological observations, such as Type Ia supernovae surveys and CMB measurements, may help distinguish between these viewpoints.

4. Integrating with Other Geometric Quantities

Ongoing work will explore the derivation of additional quantities, including:

- The Hubble parameter $H = \pi c/R$
- Vacuum energy density $\rho = \pi c^2/R$
- Planck-scale parameters expressed in terms of R, L_p, T_p, Z
- The fine-structure constant α derived from curvature and impedance relations

These extensions aim to develop a coherent geometric framework where fundamental constants emerge as interrelated rather than independently postulated.

In this way, Wave Theory aspires to provide not only reinterpretations but also concrete, testable predictions—laying out a geometric roadmap for future theoretical and observational cosmology.

Acknowledgements

The author thanks the broader physics community—both historical and contemporary—for the foundational insights that paved the way toward a more geometric understanding of nature. Particular gratitude is extended to Albert Einstein, whose interpretation of gravity as geometry remains one of the deepest truths in science. Wave Theory seeks not to overturn this insight, but to carry it forward.

This work was independently developed by the author, with conceptual and computational tools openly available from the scientific community. Computational derivations, visualizations, and theoretical checks were performed using Mathematica notebooks available through the Wave Theory GitHub repository:
`https://github.com/WaveTheoryPhysics/WaveTheory`.

Appendix: Numerical Constants Used

Table 4: Fundamental constants used in the evaluation of R (CODATA 2018).

Constant	**Symbol**	**Value**
Planck constant	h	$6.62607015 \times 10^{-34}\,\mathrm{J\,s}$
Elementary charge	e	$1.602176634 \times 10^{-19}\,\mathrm{C}$
Fine-structure constant	α	$7.2973525693 \times 10^{-3}$
Speed of light	c	$2.99792458 \times 10^{8}\,\mathrm{m/s}$
Planck length	L_p	$1.616255 \times 10^{-35}\,\mathrm{m}$
Planck time	T_p	$5.391247 \times 10^{-44}\,\mathrm{s}$
Vacuum impedance	Z	$376.730313668\,\Omega$
Hubble constant	H	$67.807848\,\mathrm{km/s/Mpc}$

Appendix: Summary of Derived Expressions for G

Below is a consolidated list of equivalent expressions for the gravitational constant G, all derived from the 3-sphere geometry and expressed through foundational physical constants. This highlights the internal coherence and multi-path derivability of G in Wave Theory.

$$
\begin{aligned}
G &= \frac{c^2}{\pi R} && \text{(Geometric form)} \\
&= \frac{c}{\pi t} && \text{(Time-based, with } t = R/c\text{)} \\
&= \frac{L_p}{\pi T_p t} && \text{(Planck units with cosmic time)} \\
&= \frac{1}{\pi \varepsilon_0 \mu_0 R} && \text{(Vacuum parameters and curvature)} \\
&= \frac{1}{\pi Z_0 t} = \frac{1}{\pi \sqrt{\varepsilon_0 \mu_0} t} && \text{(Impedance and cosmic time)}
\end{aligned}
$$

These expressions reveal the same underlying truth: G is not a coupling coefficient, but a curvature-dependent acceleration scale tied to the size and evolution of the universe itself.

Appendix: Compact Wave Theory Unit System

Table 5: Selected physical quantities in the Wave Theory unit system (compact summary).

Quantity	**SI Definition**	**WT Units**	**Interpretation**
Action	—	L^4/T	WTMedium deformation over time
Charge	A·s	$L^2/T^{1/2}$	Surface deformation per root time
Acceleration	m/s^2	L/T^2	Curvature-induced motion
Energy	J	L^3/T^2	Total WTMedium energy
Force	kg·m/s^2	L^3/T^2	Tension along curvature arc
Expansion H	1/s	$1/T$	Temporal unfolding rate
Energy dens.	J/m^3	L/T^2	Compression of WTMedium surface
Pressure	N/m^2	L/T^2	Identical to energy density

For a full extended table including additional derived quantities and detailed substitutions, see the complete Wave Theory unit system document [17].

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